

Chapter 14

Life Processes: Reproduction, Growth and Development

You had studied in class 9 that all living organisms are made up of cells which originate from pre-existing cells. You had also studied that the unfertilized eggs of animals and egg cells of plants are single cells. You know that seeds are formed from plant egg cells or ovules and germination of seeds require certain conditions like adequate water, air etc. You may also have had the experience of observing a plant growing from a small seed and an animal from a fertilized egg.

- If an unfertilized egg is a single cell, are fertilized eggs composed of several cells?
- Are seeds composed of several cells too?

You may be surprised to know that, a fertilized egg is a single cell till the process of division starts in it.

The fertilized egg starts dividing in the womb of a human female nearly 2 to 30 hours after fertilization. Divisions of fertilized eggs of hens start within 3 hours of fertilization while in several flowering plants the division starts in about 24 hours after fertilization.

- Does fertilization have any role in triggering the process of division?
- Is the role of male and female essential for the process of reproduction?
- Does reproduction produce off springs (or children) that are exact copies of their parents?
- Is reproduction essential for the life of an individual animal?
- Does growth and development have anything to do to reach reproductive phase?

We shall try to seek answers to such questions in this chapter.

All life processes go on in a coordinated and controlled manner in all living organisms. Life processes like nutrition, transport, respiration, excretion, growth and development are essential for the survival of living organisms. Reproduction gives rise to variations. Though this process is not essential for the survival of an individual living organism of a particular species, it is essential for the survival of the species as a whole.

Come let us study about the structures involved and the process of reproduction, growth and development mainly in humans (as an example of animals) and plants.

14.1 Reproduction, Growth and Development in Human beings

It is easy to distinguish between human males and females as various external characters are distinct in them. This is observed in several other organisms as well. Let us study the roles of male and female forms in the process of reproduction.

14.1.1 Reproduction: role of male and female forms

We find a sustained role of both human males and females from fertilization to organization and development of human off-springs. Such process of reproduction in which we find the role of males and females is called as sexual reproduction.

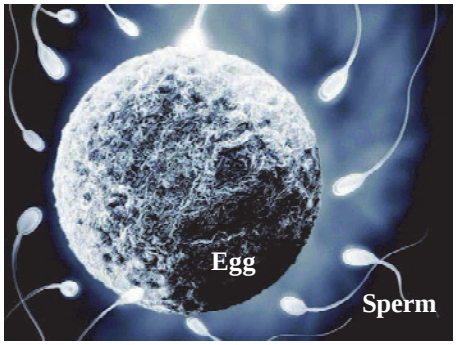


Figure- 1: Human female and male reproductive cells or gametes

As we know human life as several others, starts from a single fertilized cell. People were curious about the nature of this fertilized entity, even before the discovery of cells. They had several questions in their mind like-

- Do the male and female reproductive entities carry a small human form in them?
- Does the male entity carry all factors responsible for the development of the offspring while the female only provides for favourable conditions for the same?

For over a thousand years people have been debating over such questions. In the 17th century some scientists proclaimed that the egg itself had everything essential for life. Then, what was the role of male body and the male reproductive cell? With the development of microscopes scientists could take a closer look at microscopic structures. They did not find any small human form in these reproductive entities. They found that the male reproductive entity or sperm was small, had a nucleus and very little food resources in it. They also found that it had a long tail and it could swim upto the egg which had a nucleus surrounded by large amount of resources. Nearly 200 years ago, the proponents of cell theory, Schleiden, Schwann and Virchow made extensive studies of reproductive entities of animals and plants. They found that both the male and female reproductive cells were cells. They remarked that the role of both male and female reproductive entities were same as both were cells and contributed equally to the development of the offspring. These cells were later called as gametes (from the latin word 'gamos' meaning marriage). During this time it was also becoming clear that parental characters were passed on through these cells to the off-spring. Scientists of the time had also started proposing that the nucleus of gametes contained certain elements through which characters were passed on from one generation to the next.

It was observed that the nuclei of female and male gametes fused during fertilization. The cell thus formed (with fused male and female cells and their nuclei) was called as zygote (from Greek word that meant 'yolky'). The nucleus of the zygote contains a collection of hereditary material of parental reproductive cells. This nucleus contains information for the formation of the body of the off-spring.

We observe the role of male and female bodies in the formation of zygote in many living organisms. Distinct difference in the male and female individual bodies of several organisms is seen as in humans. The male and female bodies bear the male and female reproductive organs respectively in which reproductive cells or gametes are produced. In some other organisms like the earthworm, leech, most plants etc. the male and female reproductive organs are borne at different locations on the body of the same individual. The role of male and female can be found in several types of organisms like most vertebrates, several insects (like locusts, butterflies, houseflies, ants etc.), molluscs (snails, mussels etc.), most flowering plants, mosses, ferns etc. All such organisms reproduce sexually and role of both male and female reproductive cells is essential in such organisms. In contrast to this, in asexual form of reproduction as we see in yeast, bacteria, flatworms, several plants (like potato, grass etc.) the role of both male and female reproductive cells is not essential.

Do you know?

The generation of a type of south American lizard continues through just the females unlike most other types of lizards where both male and female forms are present. These lizards do not have male forms in any generation.



Let us study about the organization of the human male and female body and the changes that undergo in them to understand the fundamental processes that occur in the male and female bodies.

14.1.2 Growth and Development in humans

You may have observed infants growing up and wondered how a small child grows into a big individual.

Let us try to collect some data related to our body to find out about growth.

Activity-1

Find out the length of the palm and height of any child in your house or in the neighborhood as well as that of you. Carry the filled in table to class and compare the values of any of your friend (x in the table show sections you don't need to fill).

Fill in the values in the following table and compare the ratios-

Table 1: Ratios of body part

S.No.	Member	Height (H)	Length of palm (L)	H_1/H_2*	L_1/L_2*	L_1/H_1*	L_2/H_2*
1.	Small child						x
2.	You					x	

*Small child L_1, H_1, You L_2 ,H_2

- Compare yours and the small child's ratio of palm length to height obtained in the last two columns. Are they similar or different?
- What do you think this ratio might be showing?

- What does the ratio H_1/H_2 and L_1/L_2 show?
- Do you think our palm grows as fast as our height?
- Compare your ratio of palm length to height with that of your friend in school. Compare values in the last column). Is it more similar to you friend than the small child?
- On the basis of these observations can we say that our body grows in some proportion which gradually changes over time as we grow?
- Can we also say that some parts of our body grow faster than others?

Think, how many more cells are added to our height in comparison to our palm length. There are two different types of data that we collect and compare with the help of this table. One, the ratio of two different parts of our body that is, length of palm and height which indicates that parts of our body are in a particular proportion that varies slightly according to age. The other is data that gives us an idea about the rate of growth of different parts of our body.

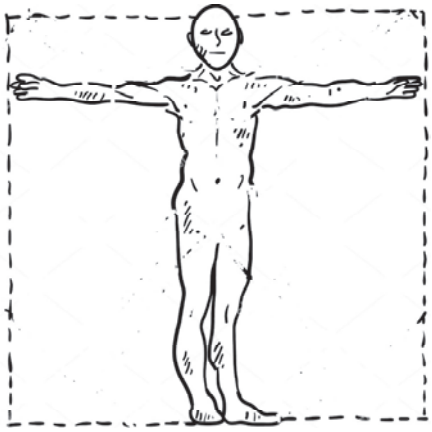


Figure-2: The human body in square

These types of observations have been made in the past as well and a drawing (like the one in Figure-2) made about 500 years back shows that our hand span and our height are same, thus a human body could fit in a square.

- Find out the values of these dimensions of your body and check if your body fits in a square as shown in the figure.

As we grow we find that certain parts of our body grow faster while certain others slowly. Gradually an adult body takes shape. During this process plenty of cells are added. Growth of our body takes place when several cells are added leading to a gradual increase of mass over time.

A graph plotted to show the change in mass of a human body over time from mothers womb to adult stage is shown in Figure 3. This thus shows growth over a particular period of time or simply the rate of growth.

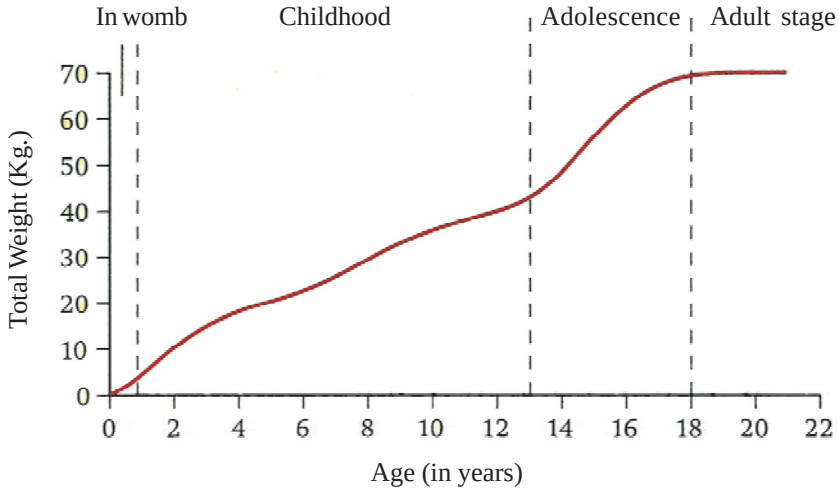


Figure- 3: Change in mass of human body (infant to adult) plotted against time

- Observe the graph and say when was growth fastest in this individual?
- What was the age after which the rate of growth slowed down?

We find that the rate of growth of various parts of our body differs. We would find that even when growth may have ceased in certain locations it may continue in others. Let us take a look at our palm.

- Do you think the growth in the area between the fingers had ceased when that in fingers continued?

Even in a child growing in the womb of mother, different parts of the body like hand, legs etc. form due to difference in rate of growth. Difference in rate of growth in different parts of our body thus leads to the development of different parts of our body.

14.1.3 Development and growth of male and female body

Development and growth brings about various changes in the body.

- What changes do you find in you now as compared to when you were in the primary section at school?

Our body changes as we grow. Around the age of 11-14 years we often find various changes taking place in our body. Along with mass there is a marked change in our height. Our reproductive organs grow and develop. Certain hormones like testosterone in males and estrogen in females are released. They stimulate functioning of reproductive parts and also aid in development and growth. Muscles grow in certain areas of our body like that on our breast (more prominently in girls), thighs etc. Hair grows in our armpits and over our reproductive areas like vagina, base of penis etc. Often acne develops on the face.

There may be behavioral changes as well. We become more conscious of our body and have several questions regarding it in our mind. When we fail to get proper answers we often show irritation. This phase of our life is called as adolescence (in Latin, it means to become mature). In human males, voice becomes low pitched, beard and moustache starts growing while in females mammary organs grow and become prominent. Changes in behavior or changes on our body are easily visible. But we hardly come to know about changes inside our body. As for example, we know that a child develops in the womb but we do not know about the changes that take place inside it and other organs of the female reproductive system in preparation for development of child. We would have to know more about our reproductive parts where internal changes take place during adolescence. See figure 4 that shows us the location of parts of female and male reproductive organs and the connections between them.

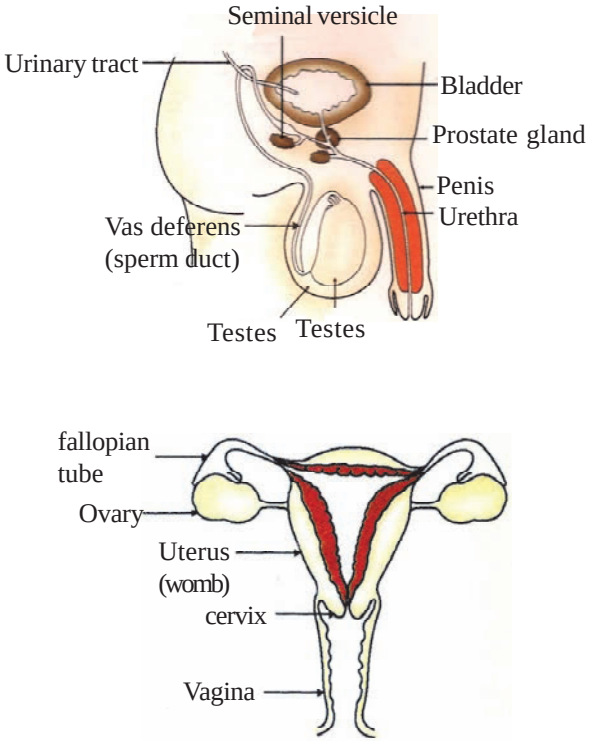


Figure- 4: The human male and female genital organs

- Observe the figure and make a list of all parts of the female reproductive system.
- Make a list of all the parts of male reproductive system.

The reproductive organs in both males and females mature during a addescence. Sperms are produced in coiled tubes inside the testes (If the coils were opened and laid out they would be around 500m long). Over 2000 lakh sperms are produced per day. Testes are found hanging in scrotal sacs outside the body. Here the temperature is lower than that of our body that helps the sperms to remain preserved.

A phase called menstrual cycle starts with the onset of adolescence is human femaly. Once every month an egg or ovum is produced by any of the ovaries. See Fig 5 for detail. Blood vessels and layers of tissue are laid in the inner wall of the uterus or the womb. This is a way in which the womb prepares for development of a child. If fertilization takes place, the development of child may continue for nearly 9 months. This phase is called as pregnancy. Menstrual cycle is arrested during pregnancy till a child is born and the body may get back to usual functioning. This is usually for a period of about 15 months from beginning of pregnancy. In case fertilization does not occur, the egg cell or ovum produced is shed off with blood and other tissues that are formed. Menstruation continues every month for nearly 35 years after reproductive maturity.

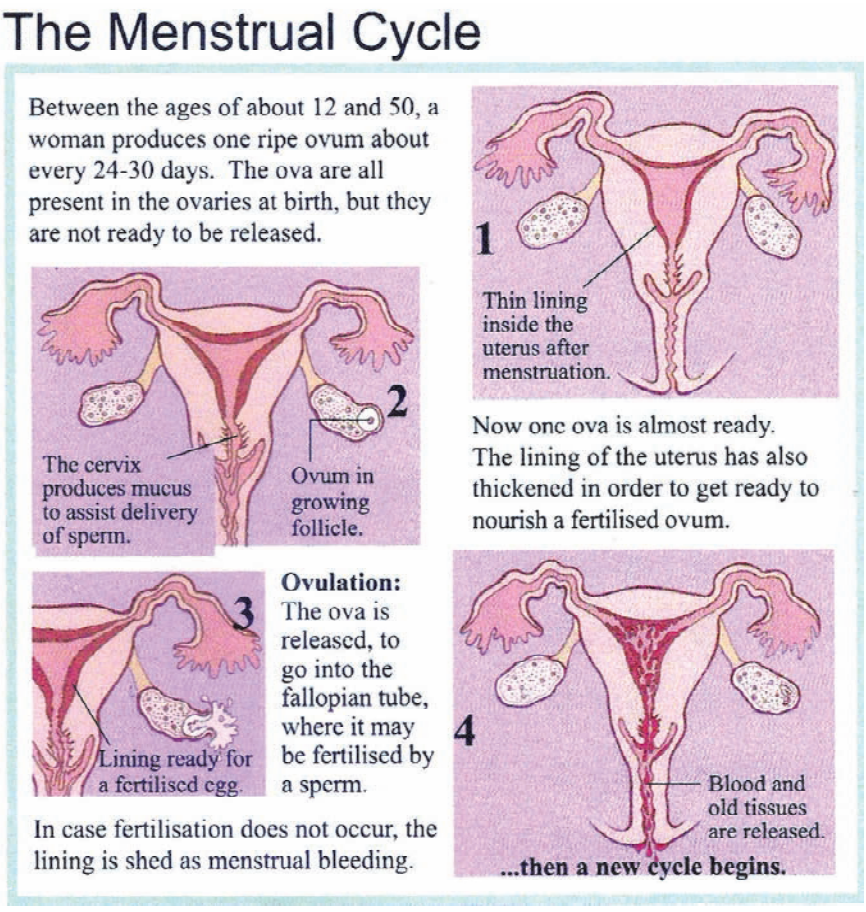


Figure- 5: Menstrual Cycle

The process of fertilization occurs via the vagina of the female. Lakhs of sperms are introduced into the vaginal passage by the penis. They travel into the fallopian tubes where the process of fertilization takes place. Only about one out of the thousands of lakhs of sperms get to meet an ovum and a successful fertilization occurs.

- But how are sperms and ovum formed?
- How are layers of cells laid in the uterus every month?
- How do we grow and how does development take place?
- We know that our body is formed of cells; do our cells increase in size or change in form when we grow?
- Do cells start dividing?

It has been observed that growth of our body is initiated mainly by division of cells as well as their increase in size. As for example in the body of a three month old child developing in the womb, four cells of the shoulder area start dividing faster than their neighboring cells and projections for hand are formed. Apart from this, it has also been observed that cells also start changing form and roles- a process we call as differentiation. As for example, cells of certain areas of the skin of a child developing in the womb are influenced by certain secretions of cells surrounding them. These skin cells transform to nerve cells under such influence.

We find similar processes of growth and development in plants as well. It is easy to observe plant growth and development in certain annual plants like pea, mustard etc. We can also observe the reproductive parts of some plants easily. Thus we shall do a few activities to study plant growth, development and reproduction.

14.2 Growth development and reproduction in plants

You have studied in earlier classes about reproductive parts of plants. Usually a flower bears the reproductive organs in plants.

Let us observe them once again

Activity-2

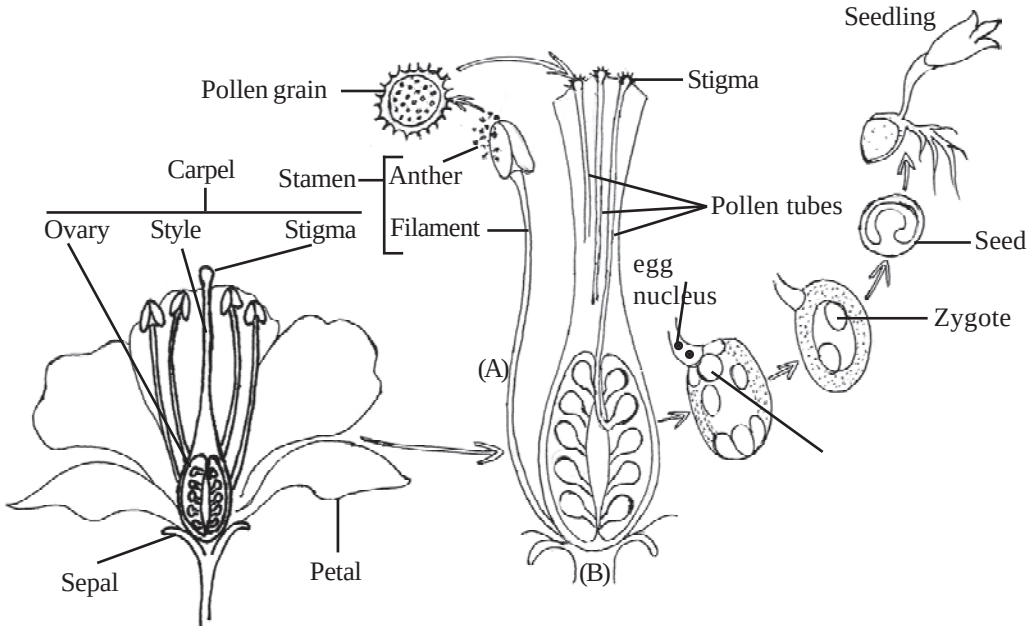
You would require a flower of dhatura or railway creeper, slide, brush, forceps, needle, petridish/watchglass, coverslip, hand lens, glycerine.

Take the flower in your hand and open it up (refer to the section of practical work for figure). Now observe the female and male reproductive parts. Figure 6 A and B shows the male and female reproductive organs of plants.

The male reproductive organ of a flower is composed of all the stamens present in the flower. Each stamen has a head formed of anther bearing pollen in pollen sacs. Pollen are structures that pass on to the stigma of the female reproductive part. The process of transfer of pollen from anther to stigma is called as pollination. If recognized on stigma they germinate to reach the ovule where the

fusion of male and female nuclei takes place. Plants may be self- pollinated when pollen of a flower reaches the stigma of the same flower or other flower on the same plant. Cross pollination occurs when pollen from flower on one plant reaches stigma of flower on another plant and is recognized there.

The female reproductive organ of a flower is composed of all the carpels. A carpel is mainly composed of three parts called as stigma, style and ovary (see figure 6). Ovules are female reproductive cells or gametes present in the ovary. Ovules (after fertilization) develop into seeds.



Stages of development fertilization to formation of seedling

Figure- 6: (A) Male reproductive part (B) Female reproductive part

Now let us observe what happens after germination of seeds. For this you would have to grow some mustard seeds in a pot and observe them regularly to see the changes that occurs in the plant as it grows.

Activity-3

Take a few mustard seeds and sow them in a medium sized pot. Note this day as day 1.Water the pot everyday with around a cup of water. Observe the pot and see when the plants grow out of the soil. Note the day. Keep observing till the plants are around 30cm in length.

1. After how many days from sowing the seed did you find the plants grow out?
2. When did you find a pair of leaves growing on the plants?
3. When did the next pair of leaves grow?
4. Was there any change in the first pair of leaves as the plant grew taller?

As in humans, growth and development occurs in plants as well. We find that leaves grow at a time after germination and flowers and fruits grow at a stage which we call as the stage of maturity of plants. The mature plant is much larger as compared to the seed from which it grows.

14.3 The relation of cell division, growth and development

The weight of a mature body is thousands of crore times more than that of a reproductive cell. Thus growth leads to a massive accumulation of resources in our body. This happens in the body of most organisms. The rate of accumulation of resources varies from one organism to another. Accumulation of resources may lead to the change in size and shape of a cell or may also lead to its division. It may also influence the cell in any other manner. In multicellular organisms, the greater the number of cells more is the resource and differentation accumulated in the body.

Let us study about cell division to know more about this.

14.3.1 Growth and Development by Cell division and Differentiation

Cell division takes place continuously in our body in processes of repair, growth, defense etc. for example layers of new cells of our skin are formed by the process of cell division.

We had studied in class 9 that new cells come from pre-existing cells. Let us try to understand how this happens.

People have been trying to find out about this long back. Studies conducted in the past 300 years have helped us to understand the process of cell division better. Scientists found that certain cells in certain tissues appeared to be dividing very fast while in some others very slowly. They also found that in certain conditions of disease, cells appeared to be growing in an uncontrolled manner (Now, we call the state of uncontrolled division of cells as cancer.). This was a challenge for and they wanted to find out what happened inside the cells. While studying division of human blood cells, a doctor named Rudolf Virchow found that the information for cell division came from the nucleus (you have studied about role of nucleus in development of Acetabularia in class 9). He found that the division of the nucleus was followed by division of cytoplasm and cell membrane.

Figure 7 A and B are types of sketches that were drawn during 1850- 1860. These are illustrations of stages of Cell Division. A, B, C, D, E shows the various stages. During this time, Walter Fleming observed the nucleus of dividing cells and found that there were thread like structures in dividing cells. He thus named the process of division 'Mitosis' (the meaning of Mitosis in Greek is thread like). He found that cells divided to produce more cells like their own. Thus production of cells or reproduction of cells occurred by the process of cell division or 'Mitosis' according to Fleming.

We know now that most cells in the body of multicellular organisms divide by mitosis to give rise to more of their kind. Several unicellular organisms like yeast, bacteria, amoeba, paramecium etc. and certain multicellular organisms like plants, hydra etc. also reproduce by processes that are based simply on mitotic cell division.

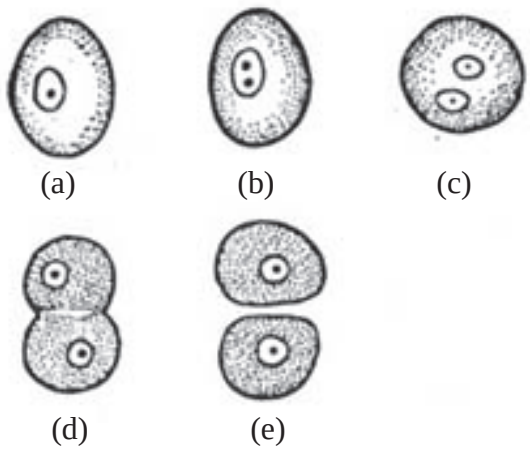


Figure- 7 (A): Cell Division

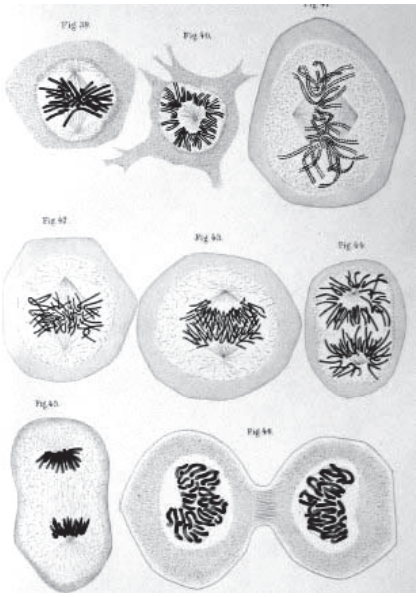


Figure-7 (B): Descriptive sketches of mitosis by Walter Fleming

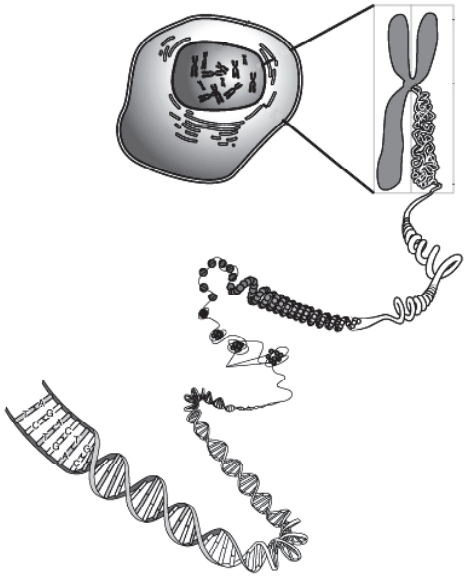
- But what is involved in starting the process of division?

Virchow had suggested that it was the nucleus. Fleming saw thread like structures in the nucleus. Scientists started observing these thread like structures to find out what they were composed off. If the cells divided and the information for division seemed to be coming from the nucleus and the thread like structures contained information even from parental generation (hereditary material) then what was their composition? Reproduction was a process that distinctly separated the living and non-living entities. Thus, the composition of the nuclear material could reveal a lot about living entities.

It was found that the thread like structures were mainly composed of proteins and a compound called deoxyribonucleic acid(DNA in short) . DNA was a material that carried information that was passed on from one generation to the next. Thus it was also called as hereditary material. Thread like DNA was wrapped around proteins to form structures that we call chromosomes. The thick thread like structures in Fig 5 are chromosomes. It was further observed that the process of mitosis lead to formation of cells that contained the same number of chromosomes as the parental cells. It is therefore also called as equational division.

As for example there are 46 chromosomes in each cell of our body. Cells formed by mitosis will also have 46 chromosomes. Mitosis occurs in most cells of our body. The process of cell division involved in growth and development of a child in the mother's womb is largely also mitosis.

Do you know?



During the decade 1940-1950 several scientists were involved in research to reveal the structure of DNA. European scientists, Rosalind Franklin and Maurice Wilkins were trying to take photographs of the DNA molecule by a special technique. In the year 1963 another European two from Cambridge, U.K. observed a photograph of DNA taken by Rosalind Franklin and made the model of DNA, the molecule of life(a molecule at the basis of cell division and differentiation). They described its structure as well. They found that it contained a type of sugar molecule, phosphate molecule and some complex compounds in a particular sequence. Watson, Crick and Wilkins received Nobel prize for their discovery. Franklin's name could not be included for the prize as she had cancer and was dead. Characters of most organisms are mainly controlled by the chemical compound DNA. The colour of the eye, skin colour etc.are all expressed due to DNA.A slight change in the chemical composition of DNA may lead to vast changes in characters. We have been able to identify areas of DNA that are responsible for specific characters and make changes in the DNA composition as well as synthesize certain small strands of the DNA molecule according to our use. Several scientists of our country like Hargobind Khorana, Lalgi Singh, Yamuna Krishnan etc. have been involved in such efforts over time. Hargobind Khorana had helped to identify specific segments of the DNA molecule that carried information for specific amino acids. He had received Nobel prize in 1968 for this discovery.

14.3.2 Life span of cell and the phase of division

Most cells of our body divide but the rate of division may be different in different types of cells. Division occuss in a very small phase in the life span of a cell. It is a very important phase for cells to maintain their existence. Fig 8 shows the extent of dividing phase in the life span of a cell. You can see phases marked as G1 and G2. These are growth 1 and growth 2 phases when resources are collected in the cell and the size of the cell increases.

- Which is the longest phase that you observe in the figure?
- Why do you think it may be the longest phase?
- Do all cells of our body divide in this manner forever?

If all cells in our body keep on dividing mitotically at the same pace or at the same time we would get a lump of cells. All cells in our body do not divide together or even at the same rate. Moreover shape and size of certain cells also change over time (faster inside the womb than outside). Thus, we see the development of different parts of our body.

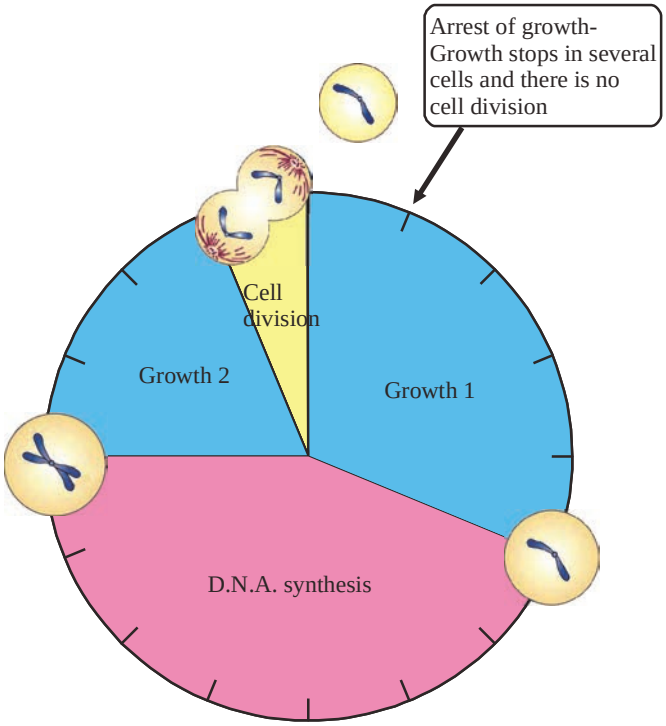


Figure-8:

14.3.3 Cell division and differentiation: and asexual reproduction, development

Most cells reproduce by the process of mitosis, and some organisms reproduce by this process of division. These may be unicellular organisms like bacteria, yeast, amoeba, paramecium etc. and multicellular organisms like hydra and plants (that grow from root shoot etc.).

The process of reproduction that takes place by the involvement of the body of a single parent organism such that cell division involved is simply mitosis, is called as asexual reproduction. There are several ways in which organisms may reproduce asexually (some examples are given in the annexure to this chapter). It is noteworthy that examples of ways in which organisms may reproduce asexually like fission or division into two parts as in yeast, budding etc. are processes that are all basically dependent on mitosis.

As we have already studied, even in our body, development and growth takes place due to different rate of mitosis in different groups of cells. Even the body of a child developing in a womb is formed by the process of mitosis. The womb itself grows as mitosis takes place in its cells. A mere 2.5-3cm long womb grows to nearly 40cm in length. Figure 9 shows some stages of development of a child in the womb.

If the egg is fertilized

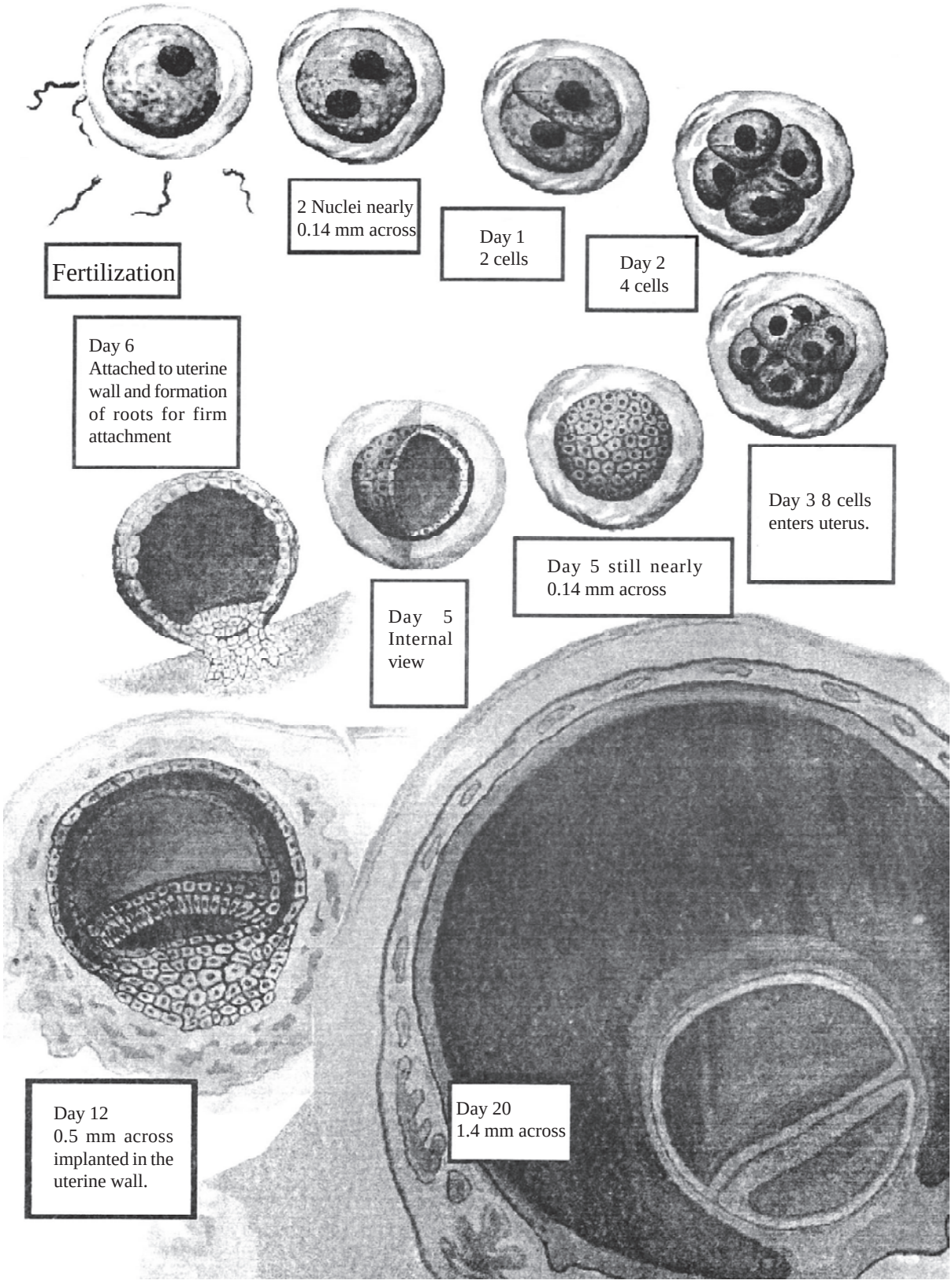


Figure- 9: Some stages of development of a child in the womb

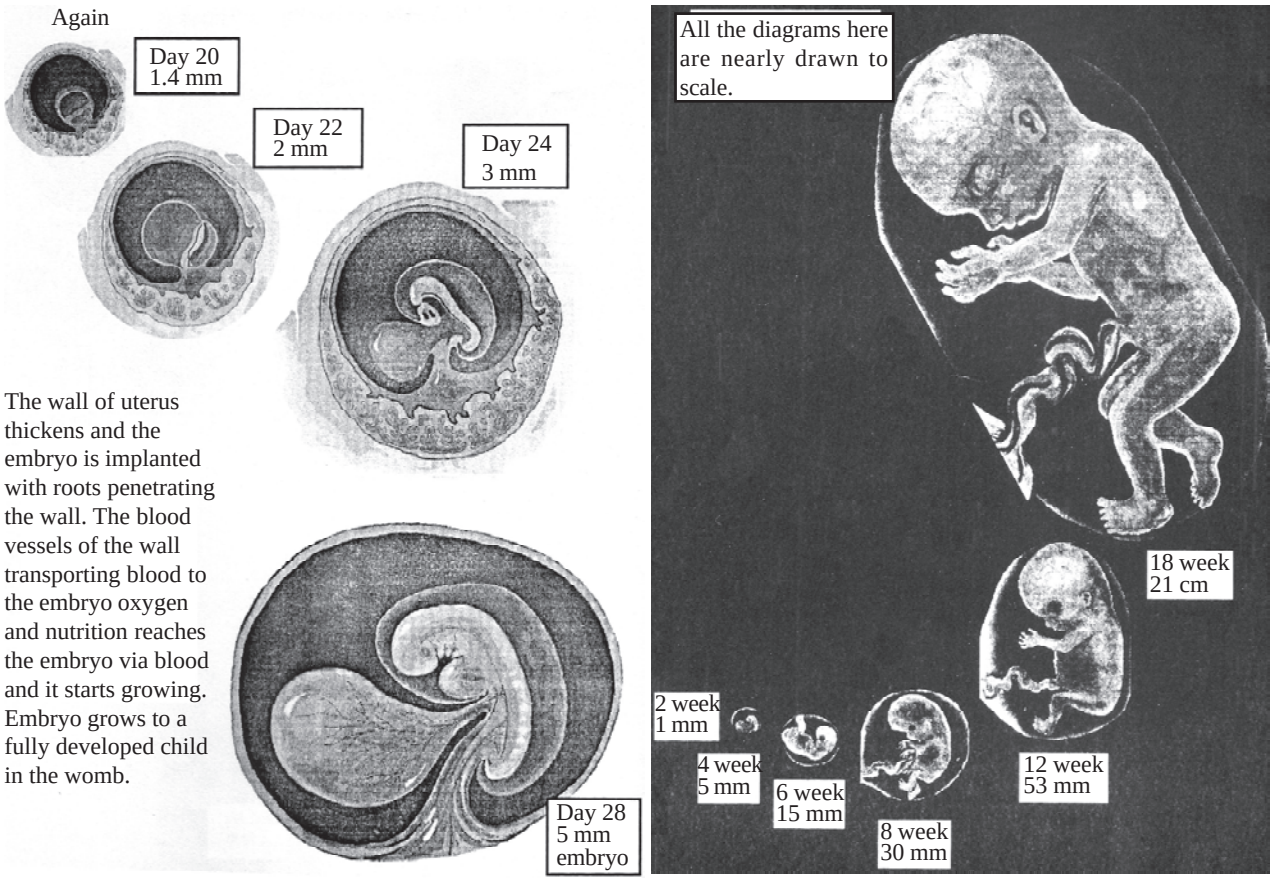


Figure- 9: Some stages of development of a child in the womb

- Observe the single cell of the first day and the ball of cells of the 5th day. Is there any change in volume?

You may see that from the first day to the fifth day there is no change in volume but the number of cells increase. It has been observed that though the volume does not change there is a marked change in mass. From the outer lining of the group of these cells of the 5th day, a connection for supply of oxygenated blood, nutrients, water etc. between the mother and child is formed. This connection is the placenta.

- We do not find any organs till the fifth day so where do life processes occur in the child?

After the fifth day a change in the shape of cells is also observed. Cells of a location may reach other locations as well. Gradually tissues, different organs and organ systems are formed and various life processes start. After the onset of cell differentiation in a fast dividing zygote, it is called as embryo (a Greek word meaning swell and grow).

- Are organs involved in the oxygenation of blood or digestion of food etc. developed and functioning in a child developing in a womb?
- Do all life processes occur in a child in the same manner as in us?
- A child remains in liquid medium in the womb so how do you think it respires?

Only after birth the lungs of a child become functional. During the period in womb of the mother, gaseous exchange occurs through the placenta as oxygenated blood reaches the body of the child from the body of the mother.

As organs and organ systems form in the womb, some of them become functional. Functional heart, eyes, limbs etc. are often seen to develop early. Reproductive organs develop in the 7th week. Certain organ systems like respiratory system, digestive system etc. develop gradually and remain non-functional till birth.

- When organs and organ systems develop and become functional at different times can they still have anything in common?
- We have studied that all cells of our body have the same DNA so how is it that cells, tissues, organs and organ systems look so different from each other?

These had been the type of questions that scientists working with DNA functioning had in mind. They found that during development, certain sections of DNA were functional in some cells while in certain other cells some other sections were functional and this brought about different changes in the cells both structurally and functionally.

- According to the discussion so far in this chapter and the chapter on evolution, would it be fair to state that different kinds of cells have the same origin?
- Would this origin be the zygote?
- Does the zygote carry information of characters of parental cells as well (remember that zygote is formed by fusion of female and male reproductive cells)?

We know that a zygote is formed after fertilization when the nuclei of male and female reproductive cells or gametes fuse. We also know that gametes are formed in the reproductive organs of our body. Similarly gametes are formed in the reproductive organs of plants and several organisms that reproduce sexually. The process of formation of gametes involves a division similar to mitosis to some extent. It differs from mitosis in such a manner that the number of chromosomes after fusion of male and female gametes remains same as that of parental cell. Let us study about the process of cell division in certain parts of the reproductive organs find out how this happens.

We have seen that organs grow and develop by mitosis but in certain areas of our reproductive organs another form of division takes place due to which gametes are formed.

14.3.4 Cell division: Sexual Reproduction and formation of gametes

We had studied that a type of cell division similar in certain aspects and differing in certain others takes place in certain parts of the male and female reproductive system. We find this in the ovaries of human female and testes of human males. Deep inside the tissues of these organs another type of cell division called meiosis takes place that leads to the formation of gametes in which the chromosome number is reduced to half as that of parental cells(See figure 10). This process of division is called as meiosis. 'Meiosis' is a Greek word that means 'being reduced' and this type of division is

also called as reductional division. As for example a human cell has 46 chromosomes while the gametes have 23 chromosomes only. Thus one chromosome of a pair in a parental cell passes to a gamete. During meiosis as gametes are formed, there is a lot of shuffling of genetic material (as compared to mitosis) between certain chromosomes.

- If gametes were formed by mitosis, how would the hereditary material increase in subsequent generations?
- What should be the amount of hereditary material in gametes, as compared to parent cells, so that the amount of hereditary material remains as in parental cells in all subsequent generations?

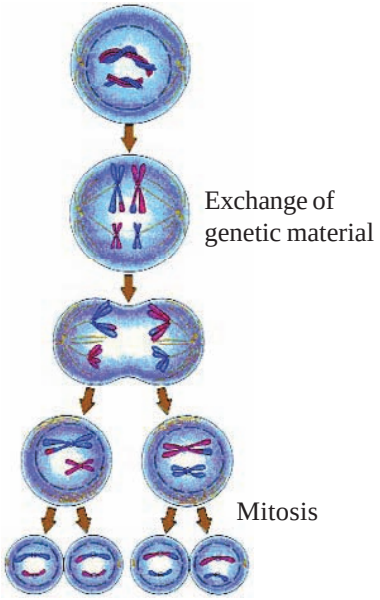
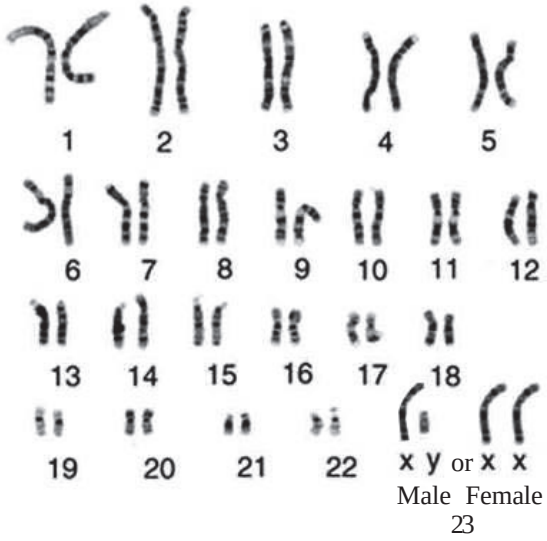


Figure- 10: Some stage of

Do you know?

Human chromosomes if arranged and displayed separately would look like the figure given here. Each cell of our body has a total of 46 chromosomes. The 23rd pair marked as XX and XY determine the sex of an individual. Those with XX being females and with XY males.



14.3.5 Asexual versus Sexual reproduction

mitosis

We found that the process of mitosis is at the base of asexual reproduction while the process of meiosis is at the base of sexual reproduction. We had stated that there is a greater shuffling of hereditary material in meiosis as compared to mitosis. Thus more shuffling of hereditary material increases the scope of variation.

- What is the role of variation in the evolution of species?
- Which form of reproduction would lead to greater scope of survival?

Since variation in a population increases the chances of adaptability to changes in the surrounding environment, therefore organisms reproducing by sexual reproduction have better chances of survival.

If sexual reproduction leads to more variations does asexual reproduction cause no variations at all? Let us do an activity involving asexual reproduction only to observe this. As we know asexual reproduction does not involve the formation of gametes. Plants reproducing by root, shoot or leaves do so asexually.

Activity-4

It would good to do this activity just after the rainy season.

Make pieces of a potato such that each one has an eye. Cut the pieces carefully such that they are blocks of similar size(let them be as closely similar as possible). Select 4-5 such blocks which are most similar. Take a large pot, fill it with soil from your garden or field and sow the blocks in at equal distances (about that of your palm length). Do not sow them too deep inside the pot.

Keep watering the pot at regular interval of time with a cup of water(at least on each alternate day).

- After how many days did you find plants growing out of the soil?
- How many plants grew? Was the number same as that of blocks?
- Were all the plants same?
- What variations did you observe in them?
- How do you think variations originated in them?
- Can we say that variations can be found even in asexually reproducing organisms as well? Why?

We found that variations though often less as compared to sexual reproduction may arise in asexually reproducing animals as well. Variations lead to the continuity of life forms on earth.

14.4 Growth development and reproduction in unicellular organisms

Most as we have studied earlier, mitosis is instrumental in the reproduction, growth and development of unicellular organisms unlike most multicellular organisms. Even in certain plants like china rose, potato etc. growth development and reproduction occurs mainly by the process of mitosis. In most multicellular organisms reproduction mainly occurs by the process of meiosis. Moreover though environmental factors influence the reproductive ability and rat of growth and development of multicellular organisms, its effects are more easily observed in unicellular organisms.

The presence of unicellular organisms in a nutrient medium is indicated by their fast reproduction and growth of the population. There may be certain factors that influence the rate of reproduction of these organisms.

Let us see the effect of changes in nutrient medium on the rate of reproduction of unicellular organism.

Activity-5

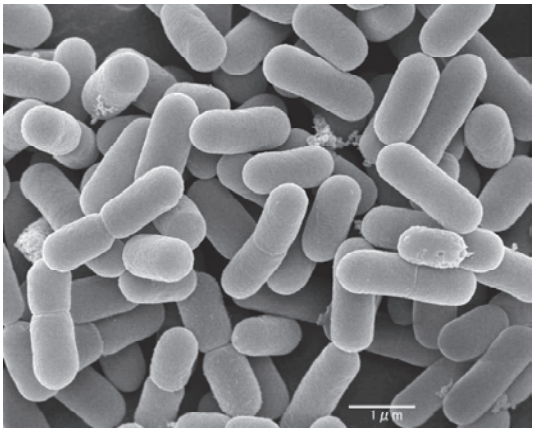
Take two bowls of same volume. Fill one bowl with warm milk which would be bowl A, while the other with boiling hot milk(pour carefully with the help of an adult) which would be bowl B. Let us add one teaspoon of curd to each of the bowls and mix it well. Now cover the bowls and leave them aside for around 5-6 hours. Remove the cover now and observe the milk in both the bowls.

- Is there any difference?
- What could be the reason for such difference?

If we observe bacteria responsible for formation of milk under an electron microscope (at around 500000 times magnification) they would look like that in Figure 10

Due to favourable conditions in bowl A most bacteria may have survived thus we find them to reproduce and curd formation is observed. Bacteria survive and grow on nutrients in milk. As their size increases they divide and their number increases.

- What do you come to know about cell division from this section?
- What is the relation between growth and reproduction according to this section?



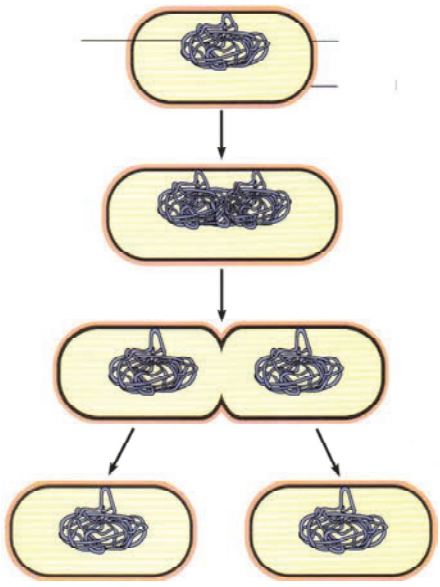
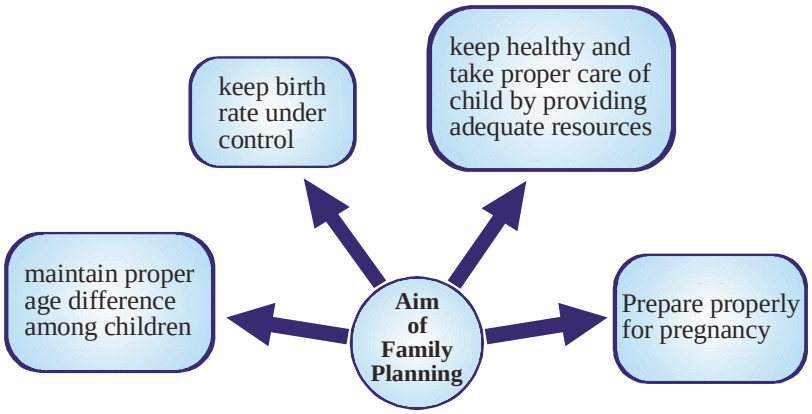


Figure-11:

14.5 Reproductive health of humans

Sexual parts are as important as all other parts of our body so we must take proper care of them by keeping them clean. We have seen, that the process of sexual maturation is gradual and takes place while general body growth is still going on. Therefore, some degree of sexual maturation does not necessarily mean that the body or the mind is ready for sexual acts or for having and bringing up children. How to decide if the body or mind is ready for this major responsibility? All of us are under



many different kinds of pressures about these issues. There can be pressure from our friends for participating in many activities, whether we really want or not. There can be pressure from families to get married and start having children. There can be pressure from government agencies to avoid having children. In this situation, making choices can become very difficult. We must also consider the possible health consequences of having sex. Our marriageable age is not same as our age of adolescence as there may be severe effect of having sex at adolescence. Neither our mind nor our body is prepared to bear its consequences at this age. Thus sex only after marriageable age (18 years for girls and 21 for boys) is advised. Even then certain constraints may be very helpful. We had discussed in class 9 that diseases can be transmitted from person to person in a variety of ways. Since the sexual act is a very intimate connection of bodies, it is not surprising that many diseases can be sexually transmitted These include bacterial infections such as gonorrhea and syphilis, and viral infections such as warts and AIDS(Acquired Immuno Deficiency Syndrome) caused by the dreaded group of HIV or human immuno-deficiency virus. Any infection that may spread through contact of body fluids like AIDS, gonorrhea or syphilis can be prevented by having sex safely. Moreover sexual act always has the potential to lead to fertilization and thereby pregnancy. Pregnancy makes major demands on the body and mind of the woman if she is not prepared for it. Thus her health may be affected adversely. There are thus many ways of avoiding pregnancy. In a country where we have a staggeringly high growth rate of population it becomes essential that we exercise ways to reduce it. All this can be achieved by having safe sex and adopting ways that would not lead to pregnancy. Such methods are called as contraceptive methods. It is advisable that human males take the initiative to exercise methods of control. Usually a literally painless operation helps to block the sperm carrying duct or vas deferens. Another means that men may adopt for safe sex is by using condoms which are coverings that fit on a penis. This arrests the passage of sperms into the vagina. There are also ways in which human females may exercise safe sex that does not lead to unwanted pregnancy. Contraception for females maybe by a minor operation of tying the fallopian tubes or blocking them. There may also be certain devices that may be introduced into the uterus to block passage of sperms to fallopian tubes, copper-T being one such device. Moreover such devices should be monitored by health practitioner regularly. There are certain medications that affect the normal release of hormones thereby arresting the release of eggs from the ovary. These pills are easily available in health centers and most widely used. But, contraceptive pills or devices for females have severe side effects and thus surgery is the best option. Of course surgery if not conducted properly or in properly sterilized (that is disease free and clean) conditions may lead to serious consequences.

Often unwanted pregnancies are aborted or removed by surgical methods. This is more of a malpractice than genuine act of birth control. Most often sex- selective abortion is conducted and a developing female child is aborted. This has lead to an alarming rate of decrease in child sex ratio in our country in spite of laws against determination of sex of developing embryo. It is important for a healthy society to have a stable sex ratio with similar female to male numbers.

Reproduction is a process that helps in increasing the population of organisms. But uncontrolled rate of increase in population will adversely affect resources. Alarming increase in population is also a challenge to raise living standards of many individuals. Birth rate and death rate of a population controls size of populations. Exercising ways and means to reduce birth rate by planning family size and having less children (around two per a married couple) would help in reducing population growth by large. As responsible citizens it is largely our duty to check population growth in this manner.

Keywords

Growth, development, mitosis, meiosis, gametes, zygote, adolescence, embryo, pregnancy, sexual reproduction, asexual reproduction



What we have learnt

- A living body grows at a definite rate.
- The increase in mass of a living body with time is growth.
- The differential rate of growth of different parts of a living body leads to development.
- During development in a multicellular body, a change in shape and size of a cell takes place along with change in shape and size of the multicellular body.
- Alike other life processes, reproduction is also an important life process that gives rise to variations in living organisms. The continuity of a species is maintained by reproduction.
- Alike sexual reproduction, asexual reproduction does not need the role of male and female. A single parent may give rise to offspring straightaway.
- An offspring receives the same amount of hereditary material(complete set) as that of the parent cell by asexual reproduction. In sexual reproduction gametes are formed that contain half the amount of hereditary material as parents. Thereafter, the offspring formed by fusion of gametes contains the same amount of hereditary material as parents.
- Asexual reproduction occurs by mitosis while gamete formation in sexual reproduction occurs by meiosis.

- Reproduction in flowering plants involves transfer of pollen grains from anther to the stigma which is referred to as pollination. This is followed by fertilization.
- Reproduction in animals involves the introduction of sperms into the female reproductive tract. This is followed by fertilization which is fusion of sperm and ovum
- Changes in the body at adolescence, such as increase in breast size in girls and new facial hair in boys, are signs of sexual maturation.
- Our health will be maintained only if we take care of all parts of our body and keep them clean. Reproductive health is extremely important and it is an integral part of our general health.
- Adopting different ways of family planning are essential for the health of a family as well as that of a society.

Assessment

1. Choose the right option-
   (i) The following type of division takes place in most cells of our body-
   (a) meiosis (b) unisexual (c) bisexual (d) mitosis
   (ii) Which of the following is not a part of the female reproductive system-
   (a) Ovary (b) Uterus (c) sperm duct (d) oviduct or fallopian tube
   (iii) Sexual reproduction leads to-
   (a) Increase in variation
   (b) Fertilization of male and female gametes lead to formation of zygote
   (c) None
   (d) Both a and b
2. Is the placenta essential for the child developing in the uterus? Why?
3. What is the role of male and female in the process of fertilization.
4. Draw a flower and label the male and female reproductive parts in it.
5. Write at least five differences between asexual and sexual reproduction.
6. What is menstruation? What is its effect on the body of a human female?
7. Write in detail about two ways of contraception.
8. Write a detail note on the reproductive processes of unicellular and multicellular organisms.
9. How does reproduction help in sustaining the population of a species?
10. How do plants that do not produce seeds reproduce?

11. What do you understand by growth and development? Describe with example.

12. What are the different aspects that need to be taken care of for the health of pregnant women?

13. You may have heard about the birth of 5 baby girls after a period of pregnancy of just 6 months. Usually a single child is born at a time in humans after pregnancy. Sometimes two children may be born. Birth of 5 children together is surprising for us. The development of the body of a child starts with the division of the zygote. Division goes on at a particular rate. Usually the birth of more than one child occurs due to the division of the zygote and the development of each division into a body of a child. Often more than one egg matures at a time and is released into the fallopian tube. The fertilization and development of all the eggs at a time gives rise to more than one child.

    Read the section and write about the ways that lead to development of more than one child in the uterus.

Annexure

There are different types of asexual reproduction. Some of them are-

(1) **Fission:** the organism divided into two or more parts. You had studied about this in activity 5 where fission in bacteria was illustrated. Apart from this we find amoeba, paramecium, malarial parasite etc. to divide in this manner.

(2) **Budding:** A bud forms due to mitosis in cells of epithelial tissue in certain organisms. The bud develops into a complete organism. This can be observed in hydra, yeast etc.

(3) **Spore formation:** in this process certain cells with a thick covering are formed which can last in adverse environmental conditions and develop into the organism with the return of favorable conditions. We usually find this in bacteria, fungi, mucor, Rhizopus etc.

(4) **Fragmentation:** Off springs are formed in planarian worms, hydra etc. by division and development of parent body into fragments containing cytoplasm and nucleus.

(5) **Vegetative propagation:** The development of organisms from parts other than reproductive system is called as vegetative propagation. You had observed in Activity-4 how plants developed from potato pieces. We see other examples like rose and grape vines developing from stem grafts, carrots, pointed and ivy gourd(parwal and kundru) by roots and begonia and bryophyllum by leaves.